

Integers

1 m
10 m
100 m
−1 m
−10 m
−100 m

Add

1 m+10 m = 11 m
1 m+10 m+(−1 m) = 10 m

Neg

−(1 m + 10 m) = 11 m
100 m+0.0015 m = 100 m
100 m×0.0015 m = 0.15 m²
(0.0015 m)³ = 3.4 × 10^{−9} m³
3.142×(1.20 cm)² = 4.52 cm²
1 m× $\frac{1\text{ m}}{1000\text{ km}}$ = 0.001 m² km^{−1}
3.14
3.14
2.0 2000 km m^{−1}

```
(  
  value: 2.0,  
  display-value: 2.0,  
  unit: (("m", 1), ("L", -1)),  
  places: 1,  
  figures: 1,  
  show: context(),  
  text: "2",  
  display: context(),  
  method: sequence(  
    context(),  
    equation(block: false, body: [+]),  
    context(),  
  ),  
  source: "add",  
  round-mode: "places",  
  is-exact: false,  
  error: 0.0,  
  error-display-value: 0.0,  
  error-text: "0",  
  error-show: context(),  
  error-display: context(),  
  error-method: equation(  
    block: false,  
    body: root(  
      radicand: sequence(  

```

```

equation(
  block: false,
  body: attach(
    base: lr(body: sequence([[], context(), []])),
    t: [2],
  ),
),
equation(block: false, body: [+]),
equation(
  block: false,
  body: attach(
    base: lr(body: sequence([[], context(), []])),
    t: [2],
  ),
),
),
),
),
)

```

6.76×10^{-4}

```

(
  value: 0.00029,
  display-value: 0.0003,
  unit: (),
  places: 4,
  figures: 1,
  show: context(),
  text: "3e-4",
  display: context(),
  method: equation(
    block: false,
    body: sequence(context(), [ ], [-], [ ], context()),
  ),
  source: "sub",
  round-mode: "figures",
  is-exact: false,
  error: 0.0,
  error-display-value: 0.0,
  error-text: "0",
  error-show: context(),
  error-display: context(),
  error-method: equation(
    block: false,
    body: root(
      radicand: sequence(
        equation(
          block: false,
          body: attach(
            base: lr(body: sequence([[], context(), []])),
            t: [2],
          ),
        ),
      ),
    ),
    equation(block: false, body: [+]),
    equation(
      block: false,

```

```

    body: attach(
      base: lr(body: sequence([(], context(), []))),
      t: [2],
    ),
  ),
),
),
),
)

```

31.41592654

Arrhenius equation is given by

$$k = Ae^{-\frac{E_a}{RT}}$$

This k , at $A = 1.50 \times 10^4 \text{ s}^{-1}$, $E_a = 5.0 \times 10^4 \text{ J mol}^{-1}$, and $T = 298 \text{ K}$, we have

$$\begin{aligned} k &= 1.50 \times 10^4 \text{ s}^{-1} \times \exp\left(\frac{-5.0 \times 10^4 \text{ J mol}^{-1}}{8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 298 \text{ K}}\right) \\ &= 2.6 \times 10^{-5} \text{ s}^{-1} \end{aligned}$$

$$\frac{1}{3.142 \text{ K}}$$

$$\frac{3 \times 10^{-4}}{9 \times 10^{-8} \text{ m}^2} \qquad -3.5$$

$$1.0$$

$$3.2 \times 10^{-4}$$

$$r_1 = \frac{0.74}{1.00} \times \exp(-0.40 \times (0.40 - (-0.80))) = 0.46$$

$$r_2 = \frac{1.00}{0.74} \times \exp(-(-0.80) \times (-0.80 - 0.40)) = 0.52$$